

Vandan Patel

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[GitHub](#) • [LinkedIn](#) • [Portfolio](#)

EDUCATION

Rutgers University–New Brunswick

Bachelor of Science in Computer Science and Finance

Bachelor of Arts in Data Science

- **GPA:** 3.94 | **Honors:** Rutgers Honors College, Dean's List (All Semesters)

New Brunswick, NJ

May 2027

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript

Technologies: React, Flask, FastAPI, Node.js, PostgreSQL, Supabase, SQLAlchemy, dbt-core, Pandas, TanStack Query

Infrastructure & Tools: AWS (EC2, S3), Docker, Git, WebSockets, REST APIs, JWT Authentication, Vite, TailwindCSS

Specialized: Financial Data Pipelines, Real-time Streaming Systems, ETL Optimization, Quantitative Analysis

WORK EXPERIENCE

Rutgers SEBS, Infrastructure & Facilities Planning

Intern, Data Analyst (Automation & Infrastructure)

New Brunswick, NJ

Sep 2025 – Present

- Eliminated 40+ annual hours of manual verification for 300+ facility nodes by engineering an automated data validation system
- Optimized data infrastructure to cut licensing costs by 60% while maintaining 100% uptime by re-architecting data infrastructure to replace legacy premium connectors with native SharePoint integrations
- Expedited data verification cycle by 35% by implementing real-time progress tracking and automated stakeholder notifications

Moweb Technologies

Intern, Data Engineering Team

Secaucus, NJ

Jul 2025 – Aug 2025

- Mitigated ETL pipeline failures from 20+ weekly incidents to under 1 within 2-week implementation by engineering null-handling logic in dbt-core using COALESCE functions and not_null data tests to prevent downstream breaks
- Reduced query latency 40% by refactoring dbt data models and optimizing Snowflake warehouse performance for client reporting
- Accelerated data reconciliation by 30% by developing LLM-assisted quality check templates, cutting manual validation overhead

PROJECTS

Algorithmic Stock Trading Bot | Python, Alpaca API, Pandas

Nov 2024 – Aug 2025

- Architected real-time monitoring system processing 50 stocks with 5-point rolling window analysis via WebSocket connections, implementing momentum-based signal detection with configurable thresholds
- Constructed dual-source pipeline handling 400+ API calls/minute with auto-failover between Yahoo Finance and Webull APIs
- Integrated risk framework with position sizing algorithms and buying power validation, enabling parameter testing across varying volatility conditions

GrindSheet – Social Fitness Analytics PWA | React, TypeScript, Supabase, TailwindCSS

May 2025 – Aug 2025

- Engineered multi-factor scoring algorithm processing 8 weighted categories for 100+ weekly logs with PR detection and ranking
- Built real-time sync with Supabase WebSockets, reducing query overhead by 80% through optimized filtering and memoization
- Launched production PWA with React 18 and TypeScript featuring offline-first architecture and centralized state management

Sunny – Live Insurance Co-Pilot (Microsoft Hackathon) | Python, FastAPI, Pathway, LLM

Jun 2025

- Deployed streaming ETL pipeline processing 100+ events from 6 live data sources with sub-2-second latency
- Devised RAG pipeline with semantic search, delivering real-time risk alerts from multi-source data for LLM recommendations
- Designed geospatial risk scoring engine with asynchronous data fetching, rate limiting, and failure recovery

LEADERSHIP AND COMPETITIONS

Rutgers Shark Tank – "PerkPal"

Business & Technical Lead, Startup Competition Finalist

Piscataway, NJ

Mar 2025

- Led cross-functional team of 3 to top 6/30 finish by conducting competitive analysis and identifying \$2.4B+ market opportunity
- Prototyped live proof-of-concept rewards aggregation platform using React and Selenium, presenting scalable technical architecture and go-to-market strategy to investor panel

Jang Star Taekwondo

Class Instructor, 1st Degree Black-Belt

Jersey City, NJ

Nov 2014 – Apr 2023

- Taught adaptive martial arts instruction across all skill levels; achieved 100% trial-to-enrollment conversion through personalized teaching methods, increasing client acquisition by 30%

ACTIVITIES & INTERESTS

Activities: Quantitative Finance Club, Rutgers Mobile App Development Club, RU Sangam, Venture Capital Club

Interests: Mountain Biking, Calisthenics, Powerlifting, Options Trading, Automotive Enthusiast